

MICROSOFT FABRIC

Deployment discovery and implementation questionnaire

A technical intake for converting and deploying the Informatica mapping as optimized
PySpark
with Microsoft SQL Server.

DEV / DEMO

m_DIM_Agreement__INTG_Agreement

English edition | Development environment | Prepared from the supplied XML and SVG

3 source definitions

2 target definitions

36 transformations

24 state variables

Start here. These facts were extracted from the supplied Informatica XML and flow diagram.

ATTRIBUTE	VALUE
Repository	RS_ACUDEV
Repository database type	Microsoft SQL Server
Folder	EDW_PRESENTATION
Mapping	m_DIM_Agreement__INTG_Agreement
Instances / connectors	32 / 357

Confirmed logical data flow

TRIGGERFabric Pipeline or manual run	→	EXECUTIONSpark Job Definition	→	TRANSFORMPySpark SCD Type 2
SECRETSAzure Key Vault	→	READDIDW + ETL Management	→	WRITEEDW DIM_Agreement

Critical connection detail

The XML points the INTG_CodeDescription lookup to EDW_DIDW02. The generated PySpark currently reads it from EDW_ETL_DW_Management. The Informatica SME and DBA must identify the authoritative Dev database before the code is deployed.

Source, target and lookup inventory - : Need following connection detail

ROLE	XML CONNECTION	OBJECT	OPERATION
Source	EDW_DIDW02	dbo.INTG_Agreement	Read
Audit source/target	EDW_ETL_DW_Management	dbo.ETL_AuditBalanceControl	Read / insert / update
Dimension target	EDW_EDW02	dbo.DIM_Agreement	Read / insert / update
Lookup	EDW_DIDW02 in XML	dbo.INTG_CodeDescription	Read
Lookup	EDW_EDW02 instance override	dbo.DIM_Agreement	Read

Provide a Dev value or confirm the conversion rule for every item in this section.

PARAMETER / VARIABLE	XML DEFAULT OR USE	DEV VALUE REQUIRED
\$\$DBSuffixDevOrQA	User mapping parameter; blank default	Exact suffix including underscore rules
\$PMFolderName	ABC filter and JobSeqID lookup	Expected folder name
\$PMWorkflowName	ABC filter and audit key	Expected workflow name
\$PMSessionName	ABC filter and audit key	Expected session name
\$PMWorkflowRunId	Audit WorkflowRunID	Fabric run ID mapping
\$PMRepositoryUserName	ETL_CreatedBy and ETL_UpdatedBy	Fabric execution identity label
\$PMCacheDir	Lookup cache path	Fabric replacement or N/A

Complete all server, database, schema, volume and permission details for the Dev environment.

LOGICALCONNECTION	SERVER FQDN	INSTANCE	PORT	ACTUAL DEV DATABASE	SCHEMA
EDW_DIDW02					
EDW_EDW02					
EDW_ETL_DW_Management					

MSSQL Environment and Connectivity

1. Where is MSSQL hosted, and are DIDW, EDW, and ETL Management on the same SQL Server instance?
2. Does SQL Server use Always On, a failover cluster, or a listener? If yes, is MultiSubnetFailover=true required?
3. Can the Fabric Spark execution account run cross-database queries across the required databases?
4. Can the complete Dev DDL be provided for all source, target, lookup, and audit objects?

Database Permissions

Does the source and target account have appropriate permission? Configure with Fabric Dev env

NOTE

We need access to the database schema or the table DDL scripts to better understand the table structures, columns, data types, keys, indexes, and constraints.

Data Processing Size

1. What are the approximate source and target row counts, database sizes, and average and maximum incremental row counts per run? This information is required to estimate the expected processing volume and configure the appropriate Fabric Spark resources.
2. What is the current Informatica execution time for this workflow, including its average, minimum, and maximum run duration?
3. What Fabric capacity SKU and Spark cores/memory are available?
4. Should the job use autoscale, a custom pool or workspace-default compute?
5. Are exact row counts required at every stage, despite the additional Spark actions?
6. Can small lookup tables be broadcast safely, and what broadcast-size limit is approved?

NOTE

We need access to the database schema or the table DDL scripts to better understand the table structures, columns, data types, keys, indexes, and constraints.

Source Database : dbo.INTG_Agreement

1. Is ETL_AuditBalanceControl.ETL_ChangeControlDT the authoritative watermark for incremental extraction?
2. Are appropriate indexes available on Agreement_DurableID, ETL_LatestUpdatedDT, ETL_RecordBeginDT, and ETL_NaturalKeyText?
3. Can duplicate rows occur for the same Agreement_DurableID and EffectiveDT? If yes, what is the deterministic tie-break rule?
4. Is ETL_DeletedDT the authoritative soft-delete indicator?
5. Can representative Dev data be provided for insert, update, delete, undelete, and rerun scenarios?

Target Database :dbo.DIM_Agreement

1. How are DIM_Agreement_ID and DIM_Agreement_DurableID generated: SQL Identity, database sequence, or Informatica MAX+1 logic?
2. What are the primary key, unique key, and foreign key definitions for dbo.DIM_Agreement?
3. Is there a unique or filtered index that guarantees only one current record for each ETL_NaturalKeyText?
4. Do any other workflows insert into or update DIM_Agreement? If yes, provide their names and schedules.

5. Does the Fabric Dev execution account have SELECT, INSERT, and UPDATE permissions on DIM_Agreement?

Audit Control

1. Does ETL_AuditBalanceControl already contain a Dev control record for this mapping?
2. Are Folder + Workflow + Session sufficient to identify an audit record, or must WorkflowRunID be included for concurrent runs?

Lookup Configuration

1. Which database is authoritative for **INTG_CodeDescription: EDW_DIDW02** or **EDW_ETL_DW_Management**?
2. Is INTG_CodeDescription.CodeDescription_ID unique? If not, what deterministic selection rule should be used?
3. What is the authoritative ORDER BY rule for the **Use Last Value** logic in LKP_DIM_Agreement?
4. Should maximum VersionID and DurableID always be calculated from EDW_EDW02.dbo.DIM_Agreement?

Concurrency Readiness

1. Can workflow concurrency be restricted to 1 during the Development demo?

Network Connectivity

1. Have DB connectivity from Fabric Spark to the SQL Server been successfully tested?
2. Is Workspace Outbound Access Protection enabled, and does it permit the required MSSQL and Key Vault connections?

Authentication and Secrets

1. Which authentication method will the Fabric job use: SQL authentication, Microsoft Entra user, service principal, or managed identity?
2. If a service principal is used, provide the Tenant ID, Client ID, authentication method, and secret or certificate expiry date.
3. Are the database credentials stored in Azure Key Vault? Provide the Key Vault URL and required secret names.
4. If Key Vault uses a private endpoint, has private connectivity from Fabric to Key Vault been configured and tested?
5. Does the Fabric execution identity have permission to retrieve the required Key Vault secrets?

Fabric Workspace and Capacity

1. What are the Fabric tenant ID, Dev workspace name, workspace ID, and region?
2. What Fabric capacity and SKU will the Dev workspace use, and will the capacity be active during the demo?
3. Does the deployment user or service principal have the required Workspace Contributor or Admin access?

Fabric Resources

4. What Lakehouse will be used? Provide its name and item ID.
5. What Fabric Environment and Spark runtime version will be used?

Execution and Configuration

6. Which execution pattern will be used: Spark Job Definition, Notebook, or Fabric Pipeline?
7. Where will environment-specific settings and Dev values be stored and injected during deployment?
8. How will the Dev Key Vault, MSSQL databases, Lakehouse, and Fabric Environment be bound to the deployment?

Deployment and Approval

9. Will the Dev deployment be performed manually or through CI/CD, and which Git provider will be used?

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SCD Type 2 and Informatica rule confirmation

PRIORITY	CONDITION	XML ACTION
1	No target natural key; first row for ENKT	INSERT new record
2	Target comparison; Type 2 differs; same EffectiveDT	UPDATE existing version
3	Target comparison; Type 2 differs; target active; incoming deleted	INSERT soft delete
4	Target comparison; Type 2 differs; different EffectiveDT	INSERT new version
5	Target active; incoming deleted	INSERT soft delete
6	Target deleted; incoming active; same EffectiveDT	UPDATE existing version
7	Target deleted; incoming active; different EffectiveDT	INSERT new version
8	Target comparison; Type 1 checksum differs	UPDATE existing version
9	Previous comparison; Type 2 differs; incoming active	INSERT new version
10	Previous comparison; incoming deleted; previous active	INSERT soft delete
11	Previous comparison; incoming and previous both deleted	skip
12	Previous comparison; incoming active after previous delete	INSERT new version
13	Previous comparison; Type 1 checksum differs	UPDATE previous version
14	All other conditions	skip

FILE	CURRENT ROLE	REQUIRED CHANGE
workflow.py	Fabric/Spark entry point	Accept deployment parameters; create Run ID; structured exit status; concurrency guard; keep readiness gate until validation passes.
m_dim_agreement__intg_agreement.py	Generated mapping	Implement ordered XML variable-port logic; correct ETL_DeleteDT; deterministic ordering; correct lookup DB; explicit EDW target.
config_m_dim_agreement__intg_agreement.yml	Connection and Spark settings	Dev database bindings; Key Vault references; encryption; timeouts; stagingschema; batch and partition settings; remove unused aliases.
fabric_target_mapping.s.json	Target projection	Validate all target fields, data types, defaults, null handling, identitycolumns and case against Dev DDL.
../fabric_runtime.py	Shared MSSQL runtime	Mapping-specific validation; Fabric authentication; mssql bulk writes; retry/deadlock handling; staging cleanup; idempotency.
conversion_status.json	Fail-closed status	Set ready=true only after Type 2 parity and end-to-end Agreement testpass.
workflow_dag.json	Dependency metadata	Verify audit begin, main writes/tallies, expiration and audit end sequencing.
README.md	Run notes	Add Dev deployment, parameters, permissions, monitoring, rollback and operations runbook.

AREA	ACCOUNTABLE OWNER
Informatica rules and parameter values	Informatica SME
MSSQL hosts, DDL and permissions	DBA
Network, private endpoints and Key Vault	Cloud / Security
Fabric workspace and deployment	Fabric Administrator
Data reconciliation	Data / Business Owner
Development demo acceptance	Project Owner

Development and deployment sequence

01	Complete this questionnaire and assign an owner to every unresolved item.
02	Receive Dev DDL, network approval, credentials, permissions and representative data.
03	Implement XML Type 2 state logic and correct source, lookup and target bindings.
04	Run static checks and focused Spark transformation tests.
05	Create or bind the Fabric Environment and default Lakehouse.
06	Configure Key Vault access and Managed Private Endpoints.
07	Deploy the package as a Spark Job Definition with Dev parameters.
08	Run read-only connection smoke tests for all three logical databases.
09	Execute one controlled run with concurrency fixed at one.
10	Reconcile every action category and audit value against Informatica.